

INFS5720 Team Project

Predicting Public Vaccination Compliance from Online Behaviour

(Due by noon **12 PM** on Tuesday 21 April 2026 via Moodle)

- This project accounts for 30% of the total marks for this course.
- The deliverables include a written project report (in pdf) and supporting code files.

Background

The COVID-19 pandemic reached Australia in early 2020. Over the following two years, the Australian Government implemented various public health measures, including travel restrictions and vaccination programs. By late 2020, COVID-19 vaccination had become a major topic of public discussion. In early 2021, as the national vaccine rollout was being planned, policymakers sought to better understand public intention toward receiving a COVID-19 vaccine. To support this effort, a UNSW research team was commissioned to conduct a survey measuring individuals' intention to be vaccinated.

Prior to the survey, the same research team had begun collecting publicly shared social media data from 1,000 consenting participants starting in late 2019. Automated tools were used to access participants' Facebook (FB) posts. All textual posts and AI-generated image descriptions were processed using automated AI-based methods.

Each post or image was analysed to estimate how likely it is related to a predefined set of topics. For each post, probabilities are assigned to topics, with only the three most relevant topics retaining non-zero probabilities. The probabilities of these three topics are normalised to sum to one.

At the beginning of 2021 (1 January 2021), approximately one year after the onset of COVID-19 in Australia, a vaccination intention survey was conducted. A total of 330 participants completed the survey, reporting their intention to receive a COVID-19 vaccine using a 4-point Likert scale (4 = strongest agreement; 1 = least agreement). These 330 labelled observations provide the training data for supervised learning.

While your first choice of model may be multiple regression, you will likely find it not working and you should think about why. You shall next change the prediction of vaccination intention into a binary classification problem. Participants with survey scores of 3 or 4 may be considered to have relatively high vaccination intention, while participants with scores of 1 or 2 may be considered to have lower vaccination intention. Another way of classification transformation can be found in the vaccination intention datafile. This transformation allows the modelling task to focus on predicting whether an individual is likely to show higher or lower vaccination intention. Note that the resulting classes may be imbalanced, meaning that one class is more frequent than the other. Therefore, model evaluation should not rely solely on prediction accuracy. You are encouraged to use evaluation measures suitable for imbalanced classification problems, such as ROC curves, AUC, lift charts, and gains charts, when comparing models.

You are provided with the following three datasets in spreadsheet format:

- **FB_post_content** – daily topic probability data derived from posts (2020 – 2021)
- **Self_report_survey**– contains basic personal information and FB usage
- **Vaccination_intention** – survey results collected on 1 January 2021

Each dataset contains a worksheet titled *Metadata*, which provides definitions for all variables.

Your task is to use the available data to build predictive models that estimate vaccination intention for the remaining 600+ participants who did not complete the survey. The objective of the model is therefore to generate a risk score or probability of high vaccination intention for each participant. These predictive insights may help policymakers design more targeted public communication strategies.

Finally, note that the number of labelled participants (330) is relatively small compared with the number of potential predictor variables provided. It is therefore unlikely that all variables will be useful in your prediction models. You will likely need to select or construct a subset of features that best capture the relevant patterns in the data. **As a result, you should expect to experiment with multiple model specifications and feature combinations during your analysis. In this project, analytical reasoning, thoughtful experimentation, and clear explanation of your modelling decisions are valued more than achieving high predictive accuracy.**

Technical Details

The key complexity of this project lies in the structure of the post-level data. There are thousands of rows record the FB posts of the 330 labelled participants. Since the prediction target is at the participant level, you must transform the post-level data into participant-level features before applying regression or neural network models.

Your first approach (**Baseline Model**) should aggregate the post-level data using simple methods such as summing or averaging topic probabilities across time.

However, this project requires you to go beyond the baseline. You will notice substantial variation in posting intensity: some participants posted thousands of times, while others posted very little or not at all. This uneven posting activity introduces modelling challenges that must be addressed in your second model (**Flexible Activity Model**).

In your Flexible Activity Model, you are expected to account for activity intensity and consider incorporating additional activity-related and time-based features. For example, you may explore (i) normalization strategies to adjust for posting frequency, (ii) measures of activity level, (iii) changes in posting behaviour over time, and (iv) temporal trends or volatility of topic probabilities. You are encouraged to justify your modelling choices clearly.

Project Pedagogy

Since this project accounts for 30% of your final grade, we adopt a structured evaluation approach to ensure fairness and clarity. While you are given freedom in your modelling choices, your presentation must answer a fixed set of questions. Each question carries a specific weight.

These questions are designed to help the marking team assess (i) your understanding of the problem, (ii) your modelling decisions, (iii) your technical implementation, (iv) your analytical reasoning, and (v) your ability to interpret and deploy predictive models.

Project Questions

Each team must answer all questions. The first three questions (Q1–Q3) focus on your understanding, design, and implementation. For Q1–Q2, describe your reasoning and modelling strategy before presenting results. Q3 focuses on model results and evaluation. Q4–Q6 assess additional effort, deployment, and reflection. Note that in this project, **you are required to implement, at least, both logistic regression and neural networks for predictions.**

Question 1 (5%) – Problem setup

- 1) For your Baseline Model, clearly define: the prediction target, the unit of analysis, and whether you treat the Likert scale as continuous or categorical (justify your choice).
- 2) Describe the data used for modelling (including time range), how you aggregated post-level data into participant-level features, and any preprocessing decisions (e.g., normalization, scaling).

Question 2 (5%) – Uneven posting and activity modelling

- 1) Explain what modelling challenges arise due to uneven posting intensity in the Baseline Model.
- 2) Describe how your Flexible Activity Model addresses this issue. What additional features did you construct? How does your approach improve upon the Baseline Model conceptually?
- 3) Discuss at least one alternative approach you considered but did not adopt.

Question 3 (45%) – Model building and results.

- 1) Build and explain your baseline model. Describe methods attempted (e.g., regression, logistic regression, neural network and their specifications). Explain your train/test protocol and evaluation metrics. Present results, including key predictors and their signs.
- 2) Repeat 1) for your Flexible Activity Model. Highlight differences in feature construction.
- 3) Compare predictive performance of both models. What improved and what did not? Discuss possible reasons.
- 4) Briefly describe other modelling attempts you explored (if any), even if they did not improve results.

Question 4 (20%) – Additional modelling efforts

- 1) Describe any additional modelling efforts, such as, feature selection, regularisation (e.g., Ridge, LASSO, etc.), alternative feature transformations, alternative neural network structures, and other preprocessing methods.
- 2) Explain what you tried and what you learned, regardless of whether performance improved.

Question 5 (15%) – Deployment to unlabelled participants

- 1) Apply your final model to the 600+ unlabelled participants. Identify the top 10% highest-risk group. Describe observable behavioural patterns within this group.

- 2) If policymakers can target only 15% of participants, propose a decision rule based on risk ranking to locate them.

Question 6 (10%) – Reflection

- 1) What limitations did you encounter in your modelling approach?
- 2) What additional data would improve predictive performance, and why?

Deliverables

The project deliverables consist of a **written project report** and supporting code files.

Project Report

Your report should begin with a brief introduction and then directly answer the six imposed project questions, clearly labelling each question (Q1–Q6) in your report.

In your answers, clearly indicate:

- The Python files used for each part of your analysis, and
- The datasets referenced in your code.

The report must:

- Be no more than **12 pages** (minimum font size 10, single line spacing),
- Be submitted in **PDF format**, and
- Follow a clear and logical structure.

Code Submission

Along with your report, you must submit a separate **zipped file** containing:

- All Python scripts used in your analysis, and
- Any processed datasets generated by your code (if applicable).

Please ensure that:

- File names are clearly labelled and consistent with those referenced in your report.
- The code runs in a logical sequence and is well organized.
- **Name and save your files using your group number, e.g., H09A_Group2.pdf and H09A_Group2.zip.**

Presentation

Each team will deliver a **10-minute presentation** during Week 10 in your tutorial session, followed by a short Q&A.

No PowerPoint slides are required. You may present directly from your PDF report, highlighting key sentences, figures, and results. This format allows greater flexibility in presenting technical details and analysis outputs.

It is recommended that you allocate your 10 minutes approximately as follows:

- Introduction, Q1 and Q2: 2 minutes
- Q3 (Model Building and Results): 4 minutes
- Q4 (Additional Efforts): 2 minutes
- Q5 and Q6 (Deployment and Reflection): 2 minutes

Please manage your time carefully. Presentations exceeding 10 minutes may be interrupted.

Marking Guideline

The project will be assessed based on the following broad criteria:

Question	Marking criteria
Q1 & Q2	Problem Understanding & Design (5x2%) • Clear definition of prediction objective and unit of analysis • Logical data preparation and feature construction • Awareness of key challenges (e.g., uneven posting)
Q3	Model Development & Evaluation (45%) • Correct implementation of baseline and flexible models • Appropriate training/testing protocol • Meaningful comparison of models • Clear explanation of results
Q4	Technical Depth & Additional Effort (20%) • Evidence of thoughtful experimentation (e.g., feature engineering, tuning, regularisation) • Demonstration of analytical effort beyond minimum requirements
Q5	Deployment & Interpretation (15%) • Proper application of final model to unlabelled participants • Logical risk ranking or targeting strategy • Insightful interpretation grounded in predictive evidence
Q6	Reflection & Communication (10%) • Recognition of limitations • Professional clarity in presentation and structure
All questions	General Expectations High-quality projects will demonstrate: • Careful reasoning and justification • Appropriate methodological choices • Clear interpretation of results • Consistency between code and report Superficial implementations or poorly justified modelling decisions will receive lower marks.